Conveyer Belt Design Report

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1. Introduction

Conveyor belt systems are essential in the modern industrial processes and they are an efficient, reliable and continuous means of moving materials through the different stages of production and handling. They have wide application in manufacturing, mining, packaging, logistics, and airports industries where large quantities of materials need to be transported safely and effectively, with minimal human intervention.

The gearbox is a vital part of a conveyor system and is the mechanical connection between the electric motor and the conveyor belt. Electric motors are usually high speed and relatively low torque, but the conveyor systems need low speed and high torque output to drive heavy loads. The gearbox is thus important in slowing the motor to the necessary speed and at the same time boosting the torque to maintain the necessary performance at different loads.

A compound reverted gear train gearbox is developed in this project to ensure the necessary reduction of speed and delivery of the torque. The gearbox is needed to transfer 5 kW of power at a motor at 1500 rpm to a conveyor belt at about 100 rpm to achieve a speed reduction ratio of about 15:1. The design entails selective and analytical designing of gears, shafts, bearings, keys, and the supporting parts to make it reliable, safe and efficient with the design life of 10 years.

The design process takes into account the basic concepts of machine design, such as the kinematic study of gear trains, the stress analysis of the mechanical parts, the choice of the material, and the real-world constraints, in terms of manufacturability, cost, and compactness. The last design target is to offer a strong and streamlined design that could be used in industrial settings.

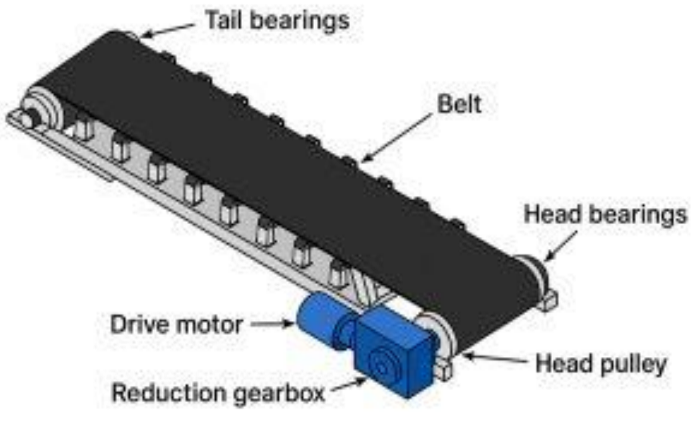

Figure 1:Components of Conveyer belt system

1.1. Scope

This project will involve the following areas:

- The work of this project will encompass the entire mechanical design and analysis of a conveyor belt system gearbox. The major aspects covered in this work are:
- A compound reverted gear train was designed to obtain the desired ratio of reduction in speed.
- Choice of suitable gear parameters such as number of teeth, module, pitch diameter and face width.
- Computation of forces on gears, tangential and radial forces.
- Gear teeth stress analysis to provide safe operation under specified loading conditions.
- Input, intermediate and output shaft design and sizing taking combined bending and torsional loading.
- Rolling element bearings selection and life analysis, as per the requirements on loads and reliability.
- Design of keyways and keys to convey torque between shafts and gears.
- Retaining elements consideration in order to position components axially.
- Design of a general gearbox system and assembly plan.

This project concentrates on analytical design and does not encompass detailed manufacturing drawings and experimental validation though possibilities of practical implementation are taken into consideration.

1.2. Objectives

The main aims of this project are as follows:

- To develop a gearbox that would be able to attain a speed reduction ratio of about 1:15.
- To provide safe and efficient transmission of 5 kW power within given operating conditions.
- To use the principles of machine design, mechanics of materials, and kinematics to apply to a real engineering problem.
- To identify appropriate gear sizes and design that meet strength and durability needs.

- To design and analyze shafts under combined loading conditions with the right factors of safety.
- In order to choose bearings that fulfil the desired life and reliability requirements.
- To develop keys and supporting elements that are designed to deliver the correct torque and integrity of the assembly.
- To come up with a small, robust and affordable gearbox design.
- To improve the knowledge of the interplay of various machine components of a whole mechanical system.

2. Design

2.1. Gearbox Design and Analysis

The gearbox is designed to achieve a speed reduction from the motor speed to the required conveyor belt speed. The required reduction ratio is 1:15, meaning the output shaft rotates at one-fifteenth of the input shaft speed. A compound reverted gear train is used to achieve this reduction efficiently while maintaining a compact design.

2.1.1. Gear Ratio and Speed Analysis

A complete stress analysis was conducted in the first phase of gear design with the module of m being chosen as 4 mm. Table 1 summarizes the results of this iteration.

Table 1:Summary of Module 4 Results

Gear	Teeth	Speed (rpm)	Load (N)	Stress (MPa)
2	20	1500	1989	41.4
3	60	500	1989	35.5
4	20	500	5969	124.3
5	100	100	5969	93.3

It was noted that the gears 2 and 3 had satisfactory performance with factors of safety of 3.6 and 4.2 respectively but the gears in the second stage (gear 4 and gear 5) had very low factors of safety. Specifically, gear 4 was approximately 1.2 factor of safety and gear number 5 was 1.6. These are lower than the acceptable design limit of 2 which means that the gear teeth are overly stressed and will likely be subjected to failure during the operating conditions.

The low factors of safety can be attributed to higher transmitted load in the second stage of the gear train and the decrease in module, which leads to reduced tooth thickness and reduces the carrying capacity of loads.

To overcome this problem, the module was raised to m = 5 mm, which raised the pitch diameter and tooth size. This led to a high decrement in the bending stress and better distribution of the loads between the gear teeth.

This was followed by a detailed design and analysis of module 5 (as shown in the next section) which showed that all gears meet the necessary strength requirements with factors of safety more than 2 against both bending and pitting failure modes.

The input speed is 1500 rpm, and the required output speed is approximately 100 rpm. Therefore, the reduction ratio is:

$$\text{Gear Ratio} = \frac{N_{input}}{N_{output}} = \frac{1500}{100} = 15$$

This corresponds to a reduction ratio of 1:15. To simplify design and avoid excessively large gears, the total ratio is divided into two stages:

- First stage ratio = 3:1
- Second stage ratio = 5:1

Thus,

$$i_1 = 3, i_2 = 5$$
$$i_{total} = i_1 \times i_2 = 3 \times 5 = 15$$

Minimum teeth must be ≥ 18 to avoid interference for 20° pressure angle.

$$N_3 = i_1 \times N_2 = 3 \times 20 = 60$$

$$N_4 = 20$$

$$N_5 = i_2 \times N_4 = 5 \times 20 = 100$$

The selected number of teeth are:

- Gear 2 = 20 teeth
- Gear 3 = 60 teeth
- Gear 4 = 20 teeth

- Gear 5 = 100 teeth

The resulting rotational speeds are:

- Gear 2: 1500 rpm
- Gear 3: 500 rpm
- Gear 4: 500 rpm
- Gear 5: 100 rpm

2.1.2. Gear Geometry

A module of 5 mm was selected based on strength and durability considerations. The corresponding pitch diameters are calculated using:

$$d = m \times N \tag{1}$$

Thus:

- Gear 2: 100 mm
- Gear 3: 300 mm
- Gear 4: 100 mm
- Gear 5: 500 mm

The face width is usually between 8-12 times of module and was selected as:

$$b = 10m = 50\,\text{mm}$$

2.1.3. Gear Force Analysis

The forces acting on the gear teeth are determined based on transmitted power. These include tangential force W_t and radial force W_r.

$$T = \frac{9550 \cdot P}{N} \tag{2}$$

- Input shaft:

$$T_2 = \frac{9550 \times 5}{1500} = 31.83\,\text{Nm}$$

- Intermediate shaft:

$$T_3 = \frac{9550 \times 5}{500} = 95.5\ \text{Nm}$$

- Output shaft (Gear 5):

$$T_5 = \frac{9550 \times 5}{100} = 477.5\ \text{Nm}$$

Tangential Force W_t is determined by equation 3

$$W_t = \frac{2T}{d} \tag{3}$$

The Tangential force on Gear pair (2–3) is given by:

$$W_{t23} = \frac{2 \times 31.83 \times 1000}{100} = 636.6\ \text{N}$$

The Tangential force on Gear pair (4–5) is given by:

$$W_{t45} = \frac{2 \times 95.5 \times 1000}{100} = 1910\ \text{N}$$

Radial Force W_r is determined by equation 4 where 20^0 is pressure angle

$$W_r = W_t \tan(20°) \tag{4}$$

$$W_r = W_t \tan\,(20°) = W_t \times 0.364$$

Radial Force W_r for Gear Pair (2–3) is given by

$$W_{r23} = 1591.5 \times 0.364 = 579\ \text{N}$$

Radial Force W_r for Gear Pair (4–5) is given by

$$W_{r45} = 4775 \times 0.364 = 1738\ \text{N}$$

Final Design Forces are given by

- Gear pair (2–3):
  - Tangential force = 1591.5 N
  - Radial force = 579 N
- Gear pair (4–5):
  - Tangential force = 4775 N
  - Radial force = 1738 N

The calculated tangential and radial forces represent the design loads acting on the gear teeth under heavy-duty operating conditions. These forces are used in the subsequent design of shafts, bearings, and gear tooth strength analysis. The results are consistent with the expected increase in load across the gearbox stages due to torque amplification.

2.1.4. Gear Tooth Strength Analysis

The gear teeth are subjected to bending stress due to the transmitted load. These stresses are evaluated using the Lewis bending equation, which provides a reliable method for estimating the strength of gear teeth. The calculated stresses are then compared with allowable material stress to determine the safety of the design.

$$\sigma = \frac{W_t}{b \cdot m \cdot Y} \tag{5}$$

Where:
W_t= design tangential load (N)
b= face width (mm)
m= module (mm)
Y= Lewis form factor

The given known values are

- Module, $\boldsymbol{m = 5}$ mm
- Face width, $\boldsymbol{b = 10m = 50}$ mm
- Pressure angle, $\boldsymbol{\phi = 20}$°
- Material: Medium Carbon Steel (C45)
- Allowable bending stress, $\boldsymbol{\sigma_{allow} = 150}$ MPa
- Service factor 2.5 already included in loads

Lewis form factor determined from standard tables:

- For 20 teeth → $\boldsymbol{Y = 0.30}$
- For 60 teeth → $\boldsymbol{Y = 0.35}$
- For 100 teeth → $\boldsymbol{Y = 0.40}$

Calculated stresses for Gear 2 (20 teeth)

$$\boldsymbol{\sigma_2 = \frac{1591.5}{50 \times 5 \times 0.30}}$$

$$\boldsymbol{\sigma_2 = \frac{1591.5}{75} = 21.22}\text{ MPa}$$

Calculated stresses for Gear 3 (60 teeth)

$$\boldsymbol{\sigma_3 = \frac{1591.5}{50 \times 5 \times 0.35}}$$

$$\boldsymbol{\sigma_3 = \frac{1591.5}{87.5} = 18.19}\text{ MPa}$$

Calculated stresses for Gear 4 (20 teeth)

$$\boldsymbol{\sigma_4 = \frac{4775}{50 \times 5 \times 0.30}}$$

$$\boldsymbol{\sigma_4 = \frac{4775}{75} = 63.67}\text{ MPa}$$

Calculated stresses Gear 5 (100 teeth)

$$\boldsymbol{\sigma_5 = \frac{4775}{50 \times 5 \times 0.40}}$$

$$\boldsymbol{\sigma_5 = \frac{4775}{100} = 47.75}\text{ MPa}$$

2.1.5. Material Selection

All gears are made of medium carbon steel (C45), which provides adequate strength and toughness for industrial applications. The allowable bending stress is approximately 150 MPa.

2.1.6. Safety Factor Evaluation

The factor of safety is calculated as:

$$n = \frac{\sigma_{allow}}{\sigma_{actual}} \tag{6}$$

The resulting factors of safety for Gear 2 are:

$$\text{FOS}_2 = \frac{150}{21.22} = 7.07$$

The resulting factors of safety for Gear 3 are:

$$\mathrm{FOS}_3 = \frac{150}{18.19} = 8.24$$

The resulting factors of safety for Gear 4 are:

$$\mathrm{FOS}_4 = \frac{150}{63.67} = 2.35$$

The resulting factors of safety for Gear 5 are:

$$\mathrm{FOS}_5 = \frac{150}{47.75} = 3.14$$

The calculated bending stresses for all gears are significantly below the allowable stress of the material. Gear 4 experiences the highest stress due to the increased torque in the second stage, resulting in the lowest factor of safety. However, the factor of safety remains above 2, which is acceptable for heavy-duty applications.

2.1.7. Gear Construction

Smaller gears (Gear 2 and Gear 4) were designed as solid discs due to their relatively small size. Larger gears (Gear 3 and Gear 5) were designed with webbed or rimmed structures to reduce weight while maintaining sufficient strength. This approach helps in minimizing material weight and inertia without compromising performance.

2.1.8. Gear Mounting

All gears are mounted on their respective shafts using parallel keys and keyways to ensure proper torque transmission. Axial positioning is achieved using shaft shoulders and retaining rings where necessary. This arrangement ensures secure attachment and prevents axial movement during operation.

2.1.9. Gear Quality

A gear quality number of $Q_v = 8$is selected, corresponding to standard commercial manufacturing processes such as hobbing. This is suitable for conveyor belt applications.

The final gear design satisfies all functional and strength requirements. The selected module and face width provide adequate load-carrying capacity, and all gears operate within safe stress limits and the minimum factor of safety meets the design requirements. The achieved gear ratio meets the required speed reduction, and the design is compact and practical for industrial implementation. The gear system designed using a module of 5 mm is safe, reliable, and efficient for the given application. All gears meet the required safety factors for bending and contact stresses. The design also incorporates weight optimization and practical mounting arrangements, making it suitable for long-term industrial use.

2.2. Shaft Design

The length of each shaft is determined based on the arrangement of components such as gears, bearings, and spacing requirements. Adequate space must be provided for mounting, assembly, and proper functioning of components.

For Shaft 1 (Gear 2 Shaft) length of shaft is determined by

- Bearing width ≈ 25 mm × 2 = 50 mm
- Gear face width = 50 mm
- Clearance (both sides) ≈ 20 mm × 2 = 40 mm

$$\text{Total Length} = 50 + 50 + 40 = 140\,mm$$

Adding margin the final size chosen was 200 mm

For Shaft 2 (Intermediate Shaft – 2 gears) length of shaft is determined by

- Bearings = 50 mm
- Two gears = 50 + 50 = 100 mm
- Spacing between gears ≈ 30 mm
- Clearances ≈ 40 mm

$$\text{Total} = 50 + 100 + 30 + 40 = 220\,mm$$

Adding margin the final size chosen was 260 mm

For Shaft 3 (Output Shaft) length of shaft is determined by

- Bearings = 50 mm
- Gear = 50 mm
- Clearance = 40 mm

$$\text{Total} = 140\,mm$$

Adding margin the final size chosen was 220 mm

Shaft lengths were determined based on the space required for bearings, gears, and assembly clearances, and were later verified against actual component dimensions

Table 2:Shaft Lengths

Shaft	**Length (mm)**
Shaft 1	200 mm
Shaft 2	260 mm
Shaft 3	220

2.2.1. Design of Shaft 1 (Gear 2 Shaft)

The forces acting on the shafts are obtained from gear analysis and resolved into vertical (tangential) and horizontal (radial) components. Each shaft is analyzed as a simply supported beam, and reactions, bending moments, and shaft diameters are determined accordingly.

- Two supports at Bearings A and B
- Gear forces act at mid-span
- Tangential force acts in vertical plane
- Radial force acts in horizontal plane

Therefore:

Two bending planes exist:

- Vertical bending (due to $\boldsymbol{W_t}$)
- Horizontal bending (due to $\boldsymbol{W_r}$)

Using the loads and lengths already determined

- $W_t = 1591.5\, N, W_r = 579\, N, L = 200\, mm$

Since the load acts at the center, reactions at both supports are equal, the free body diagram is given by

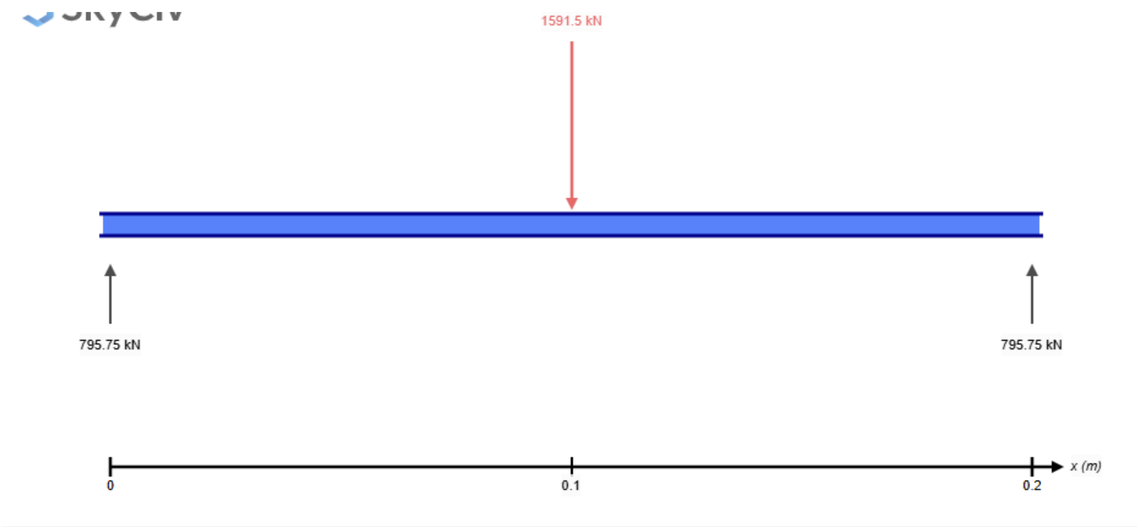

Figure 2:Free body diagram for vertical forces

Vertical reactions is determined by:

$$R_{Av} = R_{Bv} = \frac{W_t}{2} = \frac{1591.5}{2} = 795.75\ N$$

The hear force diagram for vertical plane is given by:

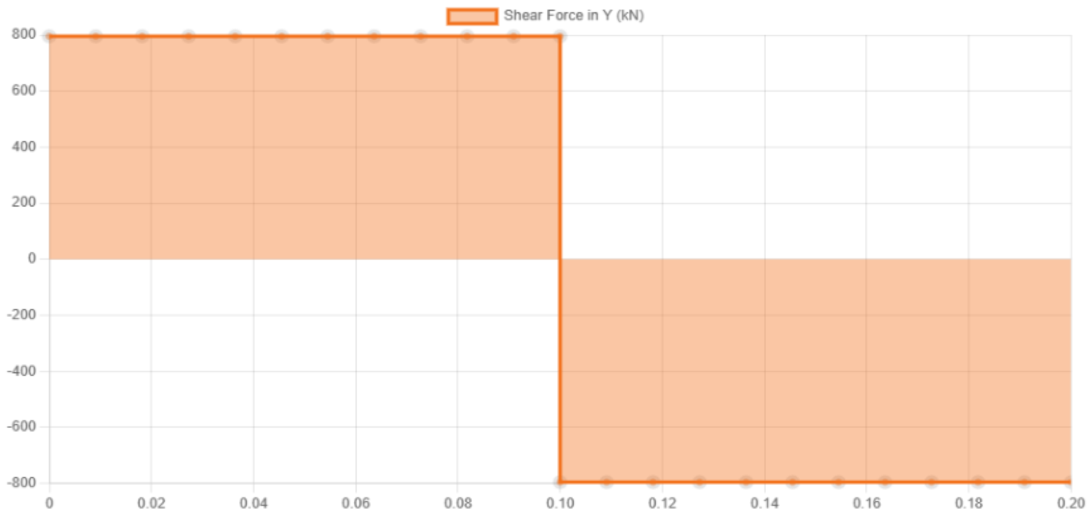

Figure 3:Shear force diagram for shaft 1 for vertical plane

Horizontal reactions is determined by:

$$R_{Ah} = R_{Bh} = \frac{W_r}{2} = \frac{579}{2} = 289.5\ N$$

The maximum bending moment occurs at the center.

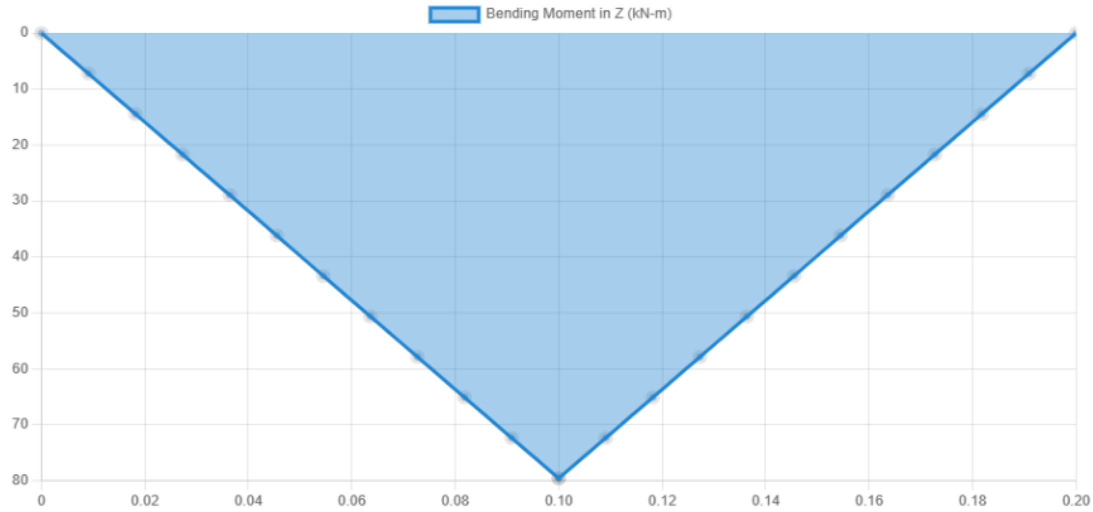

Figure 4: Bending Moment diagram for shaft 1 for vertical plane

$$M_v = \frac{W_t \cdot L}{4} = \frac{1591.5 \times 200}{4} = 79575\ Nmm$$

The deflection diagram is given by:

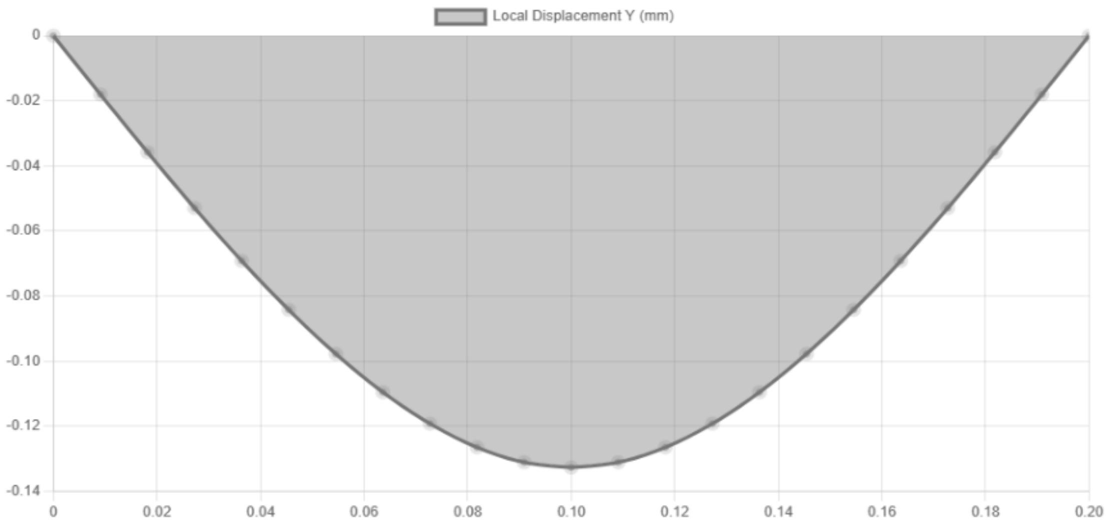

Figure 5:Deflection diagram for shaft 1 for vertical plane

Horizontal bending is determined by

$$M_h = \frac{W_r \cdot L}{4} = \frac{579 \times 200}{4} = 28950\ Nmm$$

Since bending occurs in two planes the resultant Bending Moment is given by**:**

$$M = \sqrt{M_v^2 + M_h^2} = \sqrt{(79575)^2 + (28950)^2} \approx 84600\ Nmm$$

Calculated Torque is given by

$$T = 31830\ Nmm$$

To account for combined bending and torsion equivalent Torque is determined by:

$$T_e = \sqrt{M^2 + T^2}$$

$$T_e = \sqrt{(84600)^2 + (31830)^2} \approx 90400\ Nmm$$

Using maximum shear stress theory Shaft Diameter is determined by:

$$d = \left(\frac{16T_e}{\pi\tau_{allow}}\right)^{1/3}$$

$$d = \left(\frac{16 \times 90400}{\pi \times 40}\right)^{1/3} \approx 23\ mm$$

Selected diameter = 25 mm

2.2.2. Shaft 2 (Intermediate Shaft – Two Gears)

This shaft carries two gears, so forces act at two different locations. These forces may act in opposite directions depending on rotation and must be treated carefully. The Forces is given by

- Gear 3:
  - $W_{t3} = 1591.5\ N$
  - $W_{r3} = 579\ N$
- Gear 4:
  - $W_{t4} = 4775\ N$
  - $W_{r4} = 1738\ N$
- Shaft length = 260 mm

The free body diagram is shown

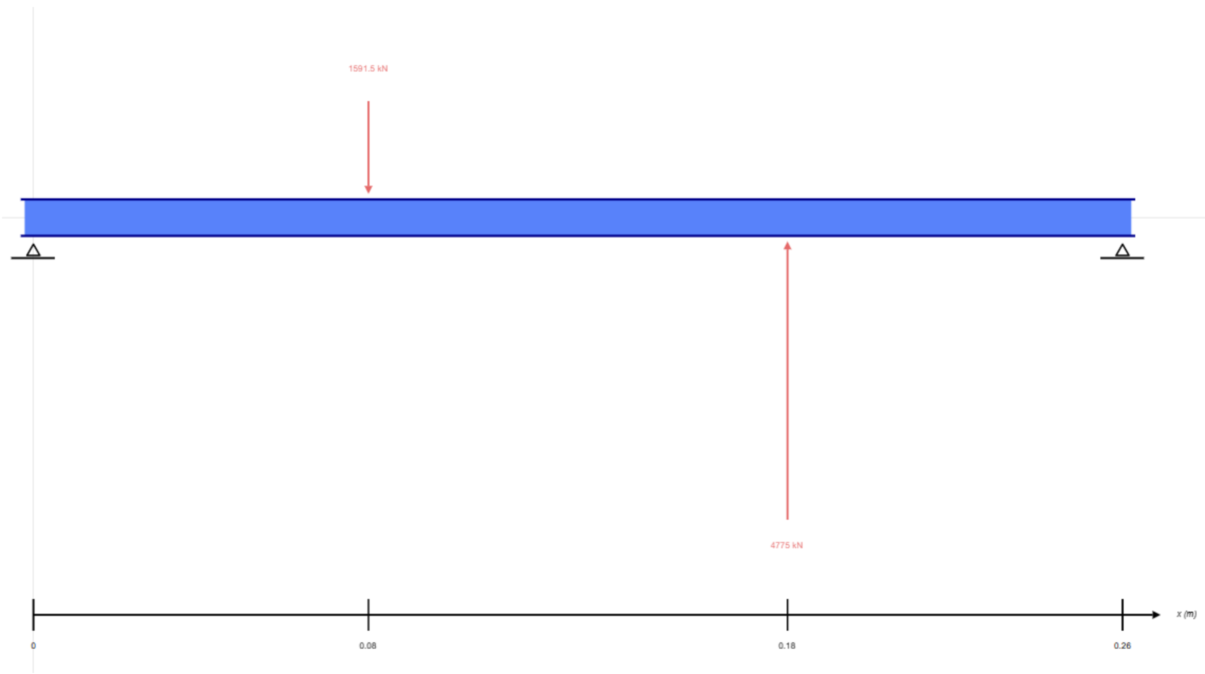

Figure 6:Free Body diagram for shaft 2

Since gears act in opposite directions: Vertical Forces is given by**:**

$$W_t = W_{t4} - W_{t3} = 4775 - 1591.5 = 3183.5\ N$$

Horizontal Forces is given by**:**

$$W_r = W_{r4} - W_{r3} = 1738 - 579 = 1159\ N$$

Vertical reactions Forces is determined by

$$R_{Av} = R_{Bv} = \frac{3183.5}{2} = 1591.75\ N$$

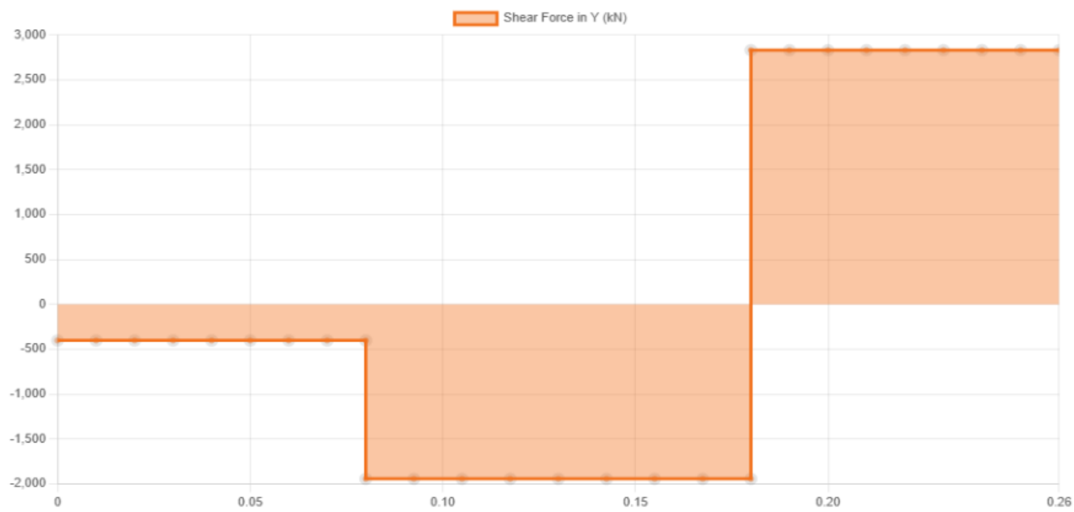

Figure 7:Shear force diagram for shaft 2 for vertical plane

Horizontal reactions Forces is determined by

$$R_{Ah} = R_{Bh} = \frac{1159}{2} = 579.5\ N$$

Bending Moments is determined by

Figure 8:Bending Moment diagram for shaft 2 for vertical plane

$$M_v = \frac{3183.5 \times 260}{4} = 206927.5\ Nmm$$

$$M_h = \frac{1159 \times 260}{4} = 75335\ Nmm$$

The deflection diagram is given by

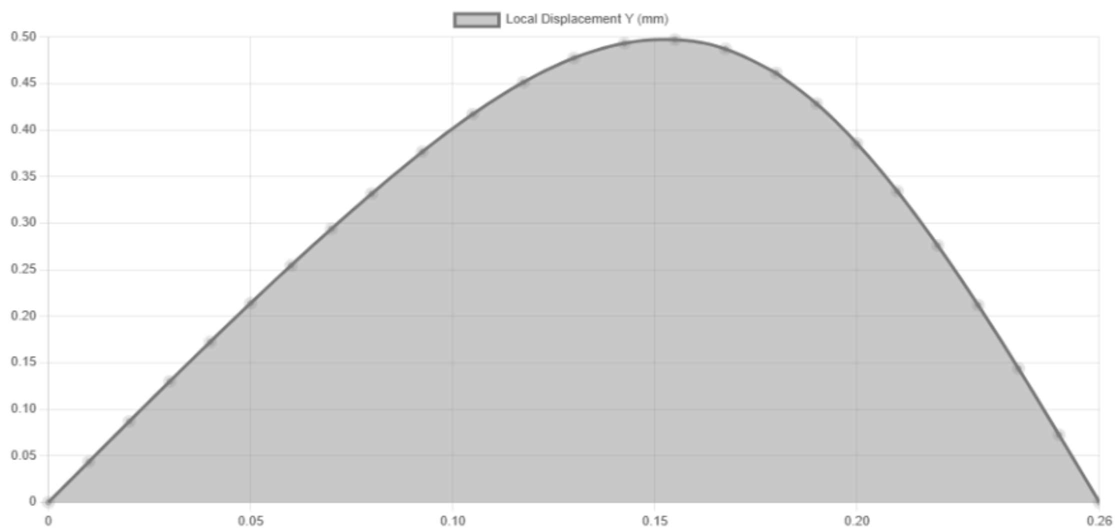

Figure 9:Deflection diagram for shaft 2 for vertical plane

Resultant Moment is determined by

$$M = \sqrt{(206927.5)^2 + (75335)^2} \approx 220000\ Nmm$$

Equivalent Torque is determined by

$$T = 95500\ Nmm$$

$$T_e = \sqrt{M^2 + T^2} \approx 240000\ Nmm$$

Diameter is determined by

$$d \approx 30\ mm$$

Selected diameter = 32 mm

2.2.3. Shaft 3 (Output Shaft – Gear 5)

- $W_t = 4775\ N$
- $W_r = 1738\ N$
- $L = 220\ mm$

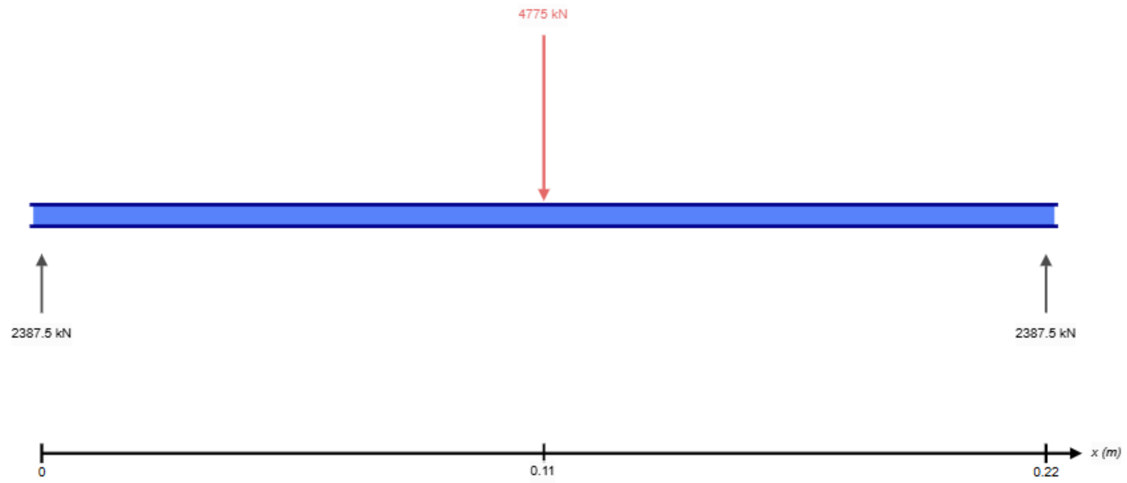

Figure 10:Free body diagram for vertical forces

Reactions is determined by

$$R_v = 2387.5\ N, R_h = 869\ N$$

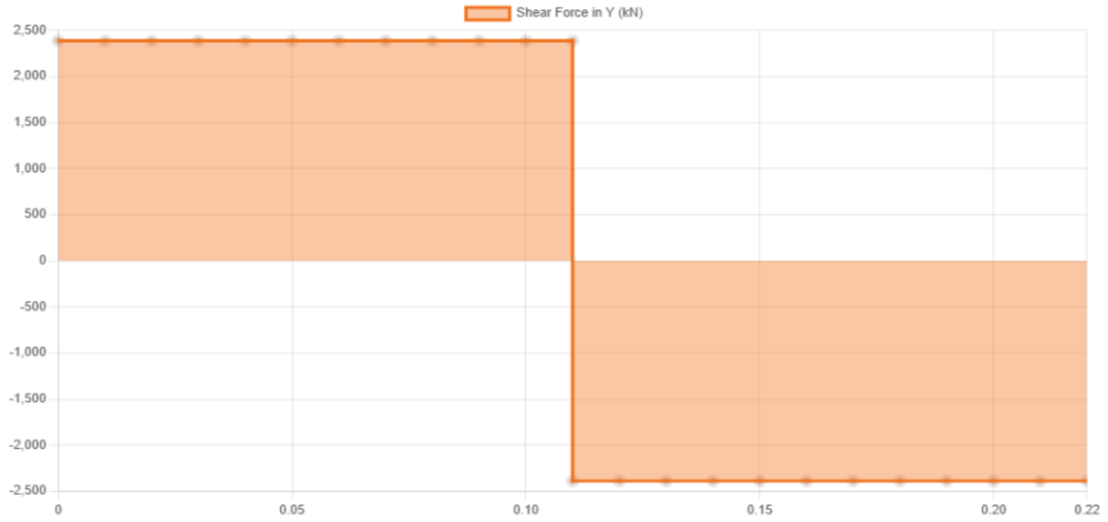

Figure 11:Shear force diagram for shaft 3 for vertical plane

Bending Moments is determined by

$$M_v = \frac{4775 \times 220}{4} = 262625\ Nmm$$

$$M_h = \frac{1738 \times 220}{4} = 95590\ Nmm$$

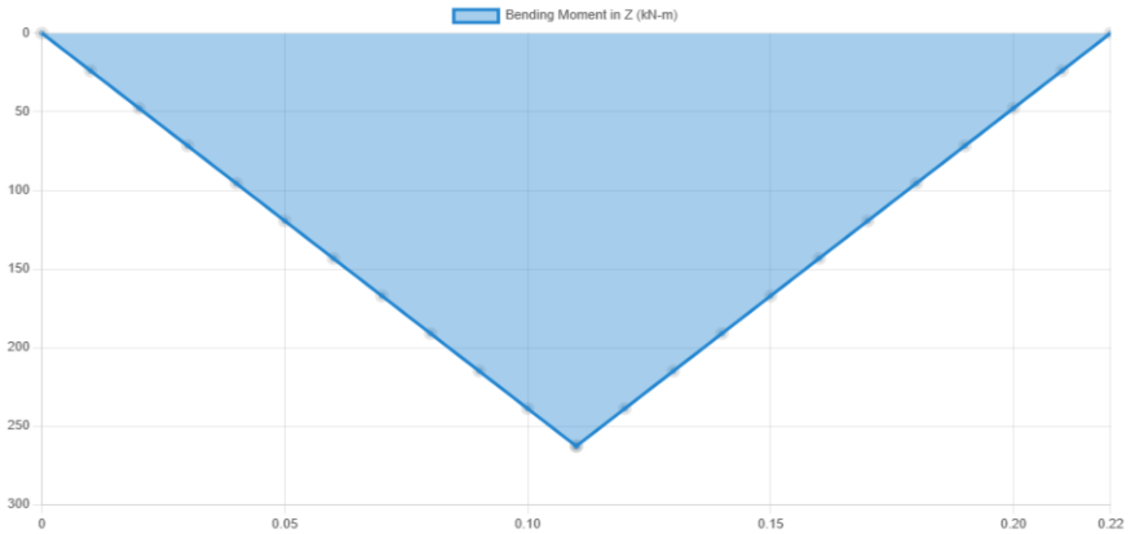

Figure 12:Bending Moment diagram for shaft 2 for vertical plane

The deflection diagram is given by

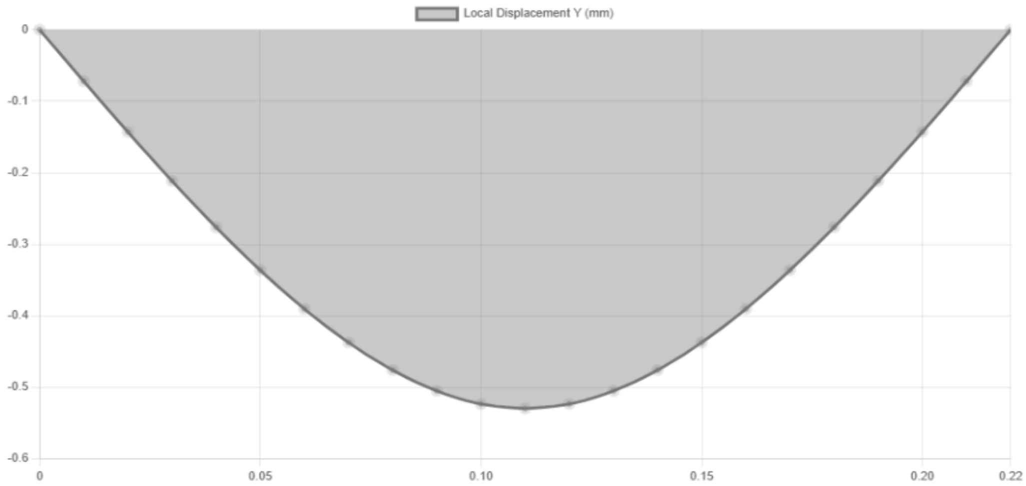

Figure 13:Deflection diagram for shaft 3 for vertical plane

Resultant Moment is determined by

$$M = \sqrt{(262625)^2 + (95590)^2} \approx 279000\ Nmm$$

Equivalent Torque is determined by

$$T = 477500\ Nmm$$

Diameter is given by

$$d \approx 38\ mm$$

Selected diameter = 40 mm

Final Shaft Dimensions is given by

Table 3:Designed Shaft Summary

Shaft	**Diameter (mm)**	**Length (mm)**
Shaft 1	25 mm	200
Shaft 2	32 mm	260
Shaft 3	40 mm	220

The shafts were analyzed under combined bending and torsion. Forces were resolved into perpendicular components and evaluated independently before combining. The resulting shaft diameters ensure safe operation under the given loading conditions with appropriate factors of safety.

2.3. Bearing Design

Bearings are selected based on applied loads, shaft speed, and required life. Deep groove ball bearings are used due to their ability to handle radial loads and moderate speeds. Reaction Forces is given by

- Shaft Reaction $= 847\ N$
- Shaft 2 Reaction $= 1700\ N$
- Shaft 3 Reaction $= 2540\ N$

Equivalent Dynamic Load is given by following Formula

$$P = a_f \times R$$

Where:

- a_f= application factor = 1.5

Equivalent Dynamic Load for Shaft 1:

$$P = 1.5 \times 847 = 1270\ N$$

Equivalent Dynamic Load for Shaft 2:

$$P = 1.5 \times 1700 = 2550\ N$$

Equivalent Dynamic Load for Shaft 3:

$$P = 1.5 \times 2540 = 3810\ N$$

Bearing Life Equation is given by following Formula

$$L_{10} = \left(\frac{C}{P}\right)^3$$

Where:

- C= dynamic load rating
- P= equivalent load

Converting Life into Hours

$$L_{10h} = \frac{L_{10} \times 10^6}{60 \times N}$$

For Shaft 1 (25 mm shaft) Selected bearing is 6205 bearing with $C = 14000\ N$

$$L_{10} = \left(\frac{14000}{1270}\right)^3 = 1331$$

$$L_{10h} = \frac{1331 \times 10^6}{60 \times 1500} = 14{,}800 \text{ hours}$$

For Shaft 2(32 mm shaft) Selected bearing is 6206 bearing with $C = 19500\ N$

$$L_{10} = \left(\frac{19500}{2550}\right)^3 = 448$$

$$L_{10h} = \frac{448 \times 10^6}{60 \times 500} = 14{,}933 \text{ hours}$$

For Shaft 3(40mm shaft) Selected bearing is 6208 bearing with $C = 31000\ N$

$$L_{10} = \left(\frac{31000}{3810}\right)^3 = 538$$

$$L_{10h} = \frac{538 \times 10^6}{60 \times 100} = 89{,}667 \text{ hours}$$

Bearings are selected for 90% reliability (L10 life) as per standard design practice.

Shaft	**Bearing**	**Life (hours)**	**Quantity**
Shaft 1	6205	14,800	2
Shaft 2	6206	14,933	2
Shaft 3	6208	89,667	2

The bearings were selected based on calculated reaction forces and operating conditions. The equivalent dynamic loads were determined using an application factor, and bearing life was evaluated using standard life equations. All selected bearings provide sufficient life and reliability for the intended operation.

2.4. Shaft Material Selection and Design Finalization

The shafts are designed to safely transmit torque while being subjected to combined bending and torsion. The bending moments are obtained from the bending moment diagrams, and torque is calculated from power transmission. A factor of safety of 2 is used, and stress concentration effects due to keyways and shoulders are considered.

For this application, medium carbon steel (C45 / AISI 1045) is selected due to:

- Good strength and toughness

- Good machinability
- Widely used in shaft applications

2.4.1. Material Properties

- Ultimate strength, $S_u = 600\ MPa$
- Yield strength, $S_y = 350\ MPa$

The allowable shear stress is calculated using factor of safety.

$$\tau_{allow} = \frac{S_y}{2 \times FOS}$$

$$\tau_{allow} = \frac{350}{2 \times 2} = 87.5\ MPa$$

Considering stress concentration:

$$\tau_{design} = 0.75 \times 87.5 = 65\ MPa$$

2.4.2. Shaft Design Using Maximum Shear Stress Theory

The shaft is subjected to combined bending and torsion. Equivalent torque is used to determine shaft diameter.

$$T_e = \sqrt{(K_b M)^2 + (K_t T)^2}$$

Where:

- $K_b = 1.5$(bending factor for shock & fatigue)
- $K_t = 1.2$(torsion factor)

For Shaft 1 following data was calculated earlier

- $M = 84600\ Nmm$
- $T = 31830\ Nmm$

Equivalent Torque is determined by

$$T_e = \sqrt{(1.5 \times 84600)^2 + (1.2 \times 31830)^2}$$
$$T_e \approx 132000\, Nmm$$

Diameter is calculated by

$$d = \left(\frac{16T_e}{\pi\tau_{design}}\right)^{1/3}$$
$$d = \left(\frac{16 \times 132000}{\pi \times 65}\right)^{1/3} \approx 26\, mm$$

Selected Diameter was 25 mm so modified to 26mm to incorporate fatigue as well.

For Shaft 2 following data was calculated earlier

- $M = 220000\, Nmm$
- $T = 95500\, Nmm$

Equivalent Torque is determined by

$$T_e = \sqrt{(1.5 \times 220000)^2 + (1.2 \times 95500)^2}$$
$$T_e \approx 350000\, Nmm$$

Diameter calculated by

$$d \approx 32\, mm$$

Already Selected Diameter= 32 mm good design estimation

For Shaft 3 following data was calculated earlier

- $M = 279000\, Nmm$
- $T = 477500\, Nmm$

Equivalent Torque is determined by

$$T_e = \sqrt{(1.5 \times 279000)^2 + (1.2 \times 477500)^2}$$
$$T_e \approx 710000\, Nmm$$

Diameter calculated by

$$d \approx 40\, mm$$

Already Selected Diameter= 40 mm good design estimation

The shaft diameters are varied along their length to accommodate bearings, gears, and keys, resulting in stepped shafts with shoulders.

2.5. Key and Keyway Design

Keys are used to transmit torque between shaft and gear. The design ensures safety against shear and crushing failure. Selected Material is Mild Steel

- Allowable shear stress: $\tau_k = 40\ MPa$
- Allowable crushing stress: $\sigma_c = 80\ MPa$

Table 4:Key Dimensions (Standard Proportions)

Shaft Diameter	Key Width (b)	Key Height (h)
25 mm	8 mm	7 mm
32 mm	10 mm	8 mm
40 mm	12 mm	8 mm

(A) Shear Failure

Key resists torque through shear along its cross-section.

$$T = \tau_k \cdot b \cdot L \cdot \frac{d}{2}$$

For Shaft 3

$$477500 = 40 \times 12 \times L \times 20$$

$$L = \frac{477500}{9600} \approx 50\, mm$$

(B) Crushing Failure

Key is compressed between shaft and hub.

$$T = \sigma_c \cdot \frac{h}{2} \cdot L \cdot \frac{d}{2}$$

Calculation

$$477500 = 80 \times 4 \times L \times 20$$

$$L = \frac{477500}{6400} \approx 75\, mm$$

Shaft	Key Size (mm)	Length (mm)
Shaft 1	8 × 7	40
Shaft 2	10 × 8	55
Shaft 3	12 × 8	75

The shafts were designed using maximum shear stress theory with appropriate factors for shock and fatigue. Stress concentration due to keyways and shoulders was accounted for by reducing allowable stress. Keys were designed to safely transmit torque without failure in shear or crushing. Keys are designed as sacrificial elements and act as weak links, ensuring failure occurs in the key rather than costly components like shafts or gears

2.6. Retaining Ring Design and Calculation

Retaining rings (snap rings) are used to prevent axial movement of components such as bearings and gears along the shaft. They are seated in grooves machined on the shaft and must be capable of resisting axial thrust generated during operation. The retaining ring must safely resist:

- Axial force due to gear thrust
- Assembly forces
- Shock loading

A factor of safety of 2 is assumed.

Axial Thrust Force Calculation

For spur gears, axial force is typically negligible, but small axial loads may arise due to:

- Misalignment
- Manufacturing errors
- Bearing reactions

A conservative assumption is made:

$$F_a = 0.1 \times W_t$$

For Shaft 1

$$F_a = 0.1 \times 1591.5 = 159\, N$$

For Shaft 2

$$F_a = 0.1 \times 3183.5 = 318\, N$$

For Shaft 3

$$F_a = 0.1 \times 4775 = 478\, N$$

The retaining ring resists axial load through shear along its cross-section.

$$F = \tau \cdot A$$

Where:

- τ= allowable shear stress
- $A = t \cdot w$= cross-sectional area
- Material: Spring steel
- Allowable shear stress:

$$\tau = 300\, MPa$$

Table 5:Retainng ring Design Summary

Shaft	Thickness (t)	Width (w)
26 mm	2 mm	3 mm
32 mm	2.5 mm	4 mm
40 mm	3 mm	5 mm

Strength Calculations for retaining ring is given by

For Shaft 1

$$A = 2 \times 3 = 6\, mm^2$$

$$F_{max} = 300 \times 6 = 1800\, N$$

$$FOS = \frac{1800}{159} \approx 11.3$$

For Shaft 2

$$A = 2.5 \times 4 = 10\, mm^2$$

$$F_{max} = 300 \times 10 = 3000\, N$$

$$FOS = \frac{3000}{318} \approx 9.4$$

For Shaft 3

$$A = 3 \times 5 = 15\,mm^2$$

$$F_{max} = 300 \times 15 = 4500\,N$$

$$FOS = \frac{4500}{478} \approx 9.4$$

Grooves must be machined to properly seat the retaining ring.

Table 6:retaining Ring Grooves

Shaft	Groove Depth	Groove Width
26 mm	1 mm	3 mm
32 mm	1.2 mm	4 mm
40 mm	1.5 mm	5 mm

The retaining rings were designed to withstand axial loads with high factors of safety. The selected dimensions ensure secure axial positioning of components while maintaining ease of assembly and disassembly.

3. Layout and CAD Model

The gearbox designed was modeled in SolidWorks

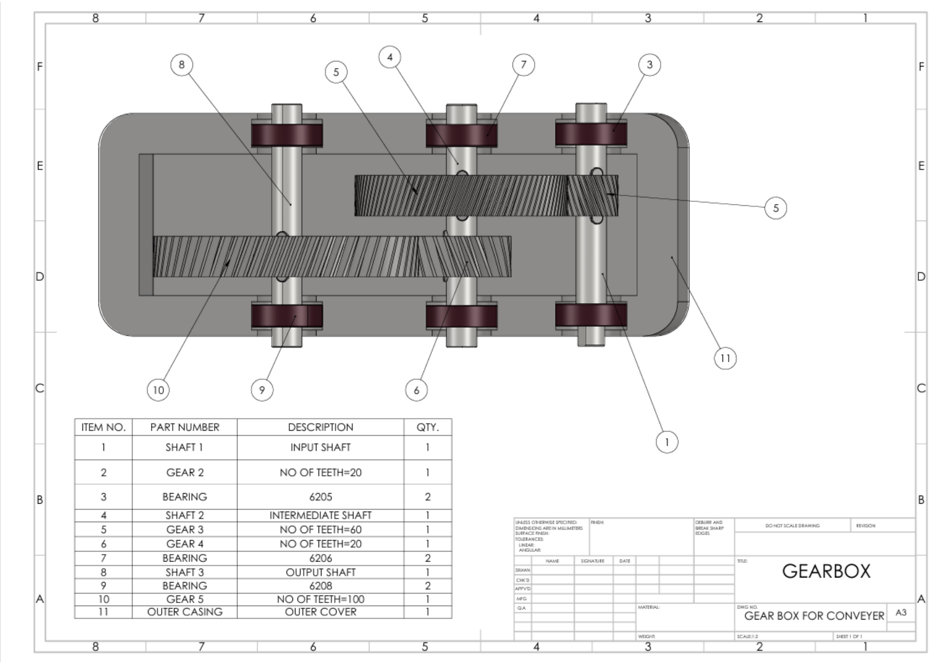

Figure 14:Layout of designed gearbox

The assembled CAD model is designed in SolidWorks

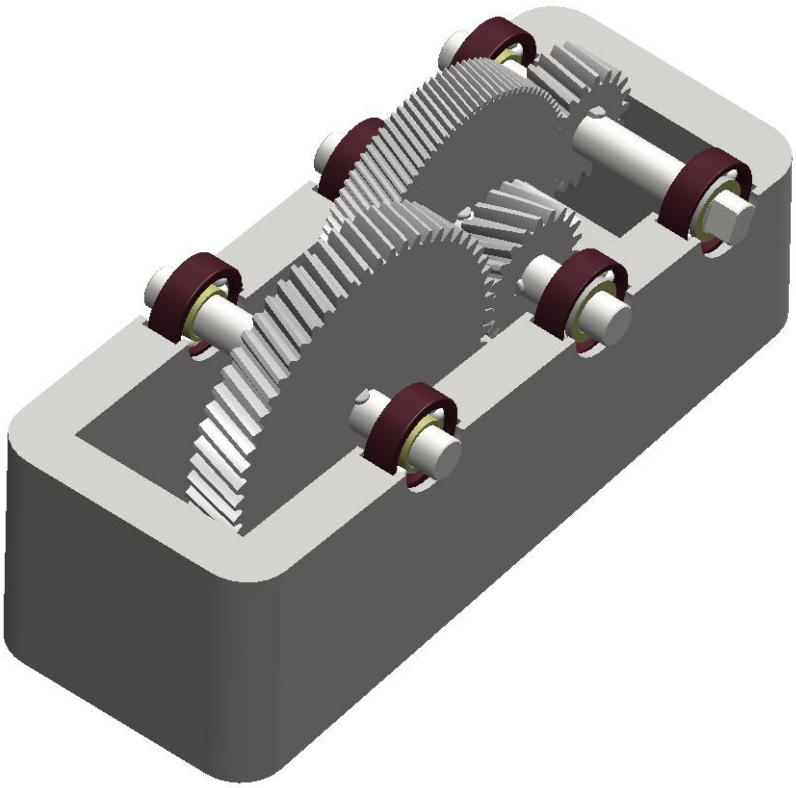

Figure 15:Assembled CAD model

The drawing of shaft 1 is shown below

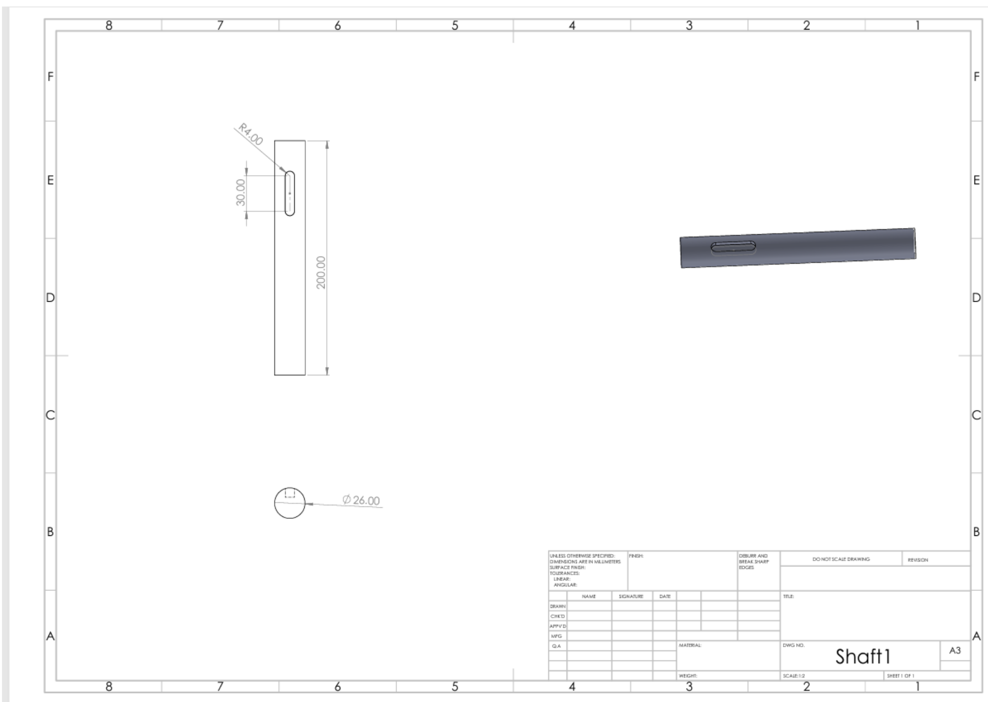

Figure 16:Designed shaft 1

The drawing of shaft 2 is shown below

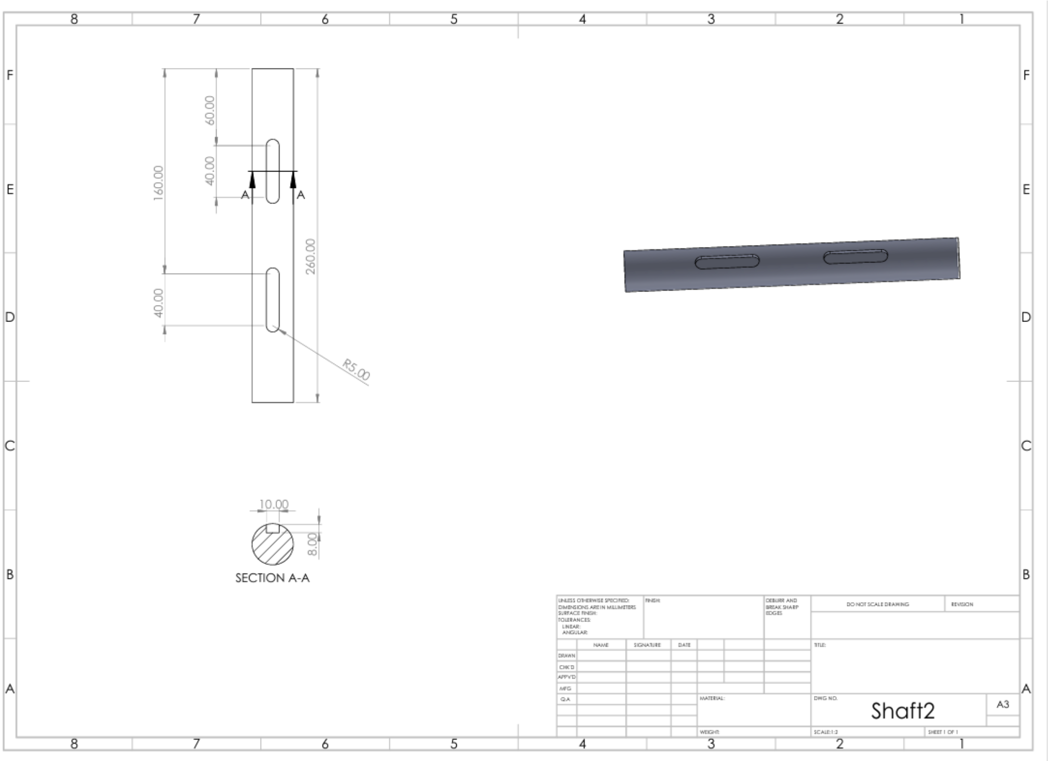

Figure 17:Designed shaft 2

The drawing of shaft 3 is shown below

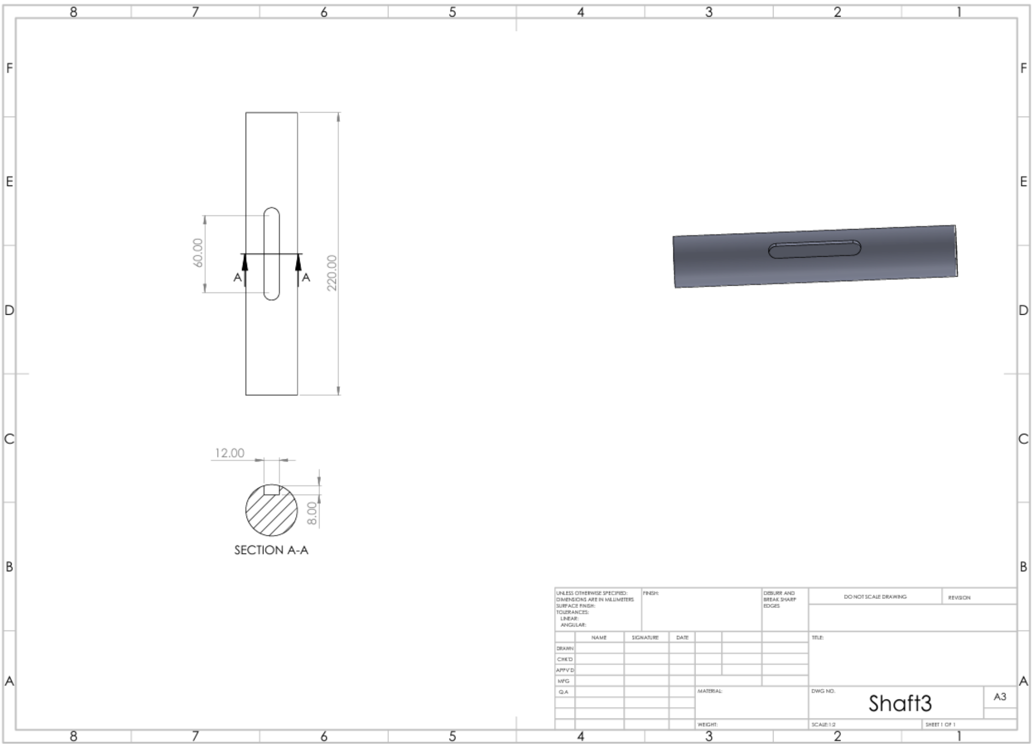

Figure 18:Designed shaft 3

The drawing of key 1 is shown below

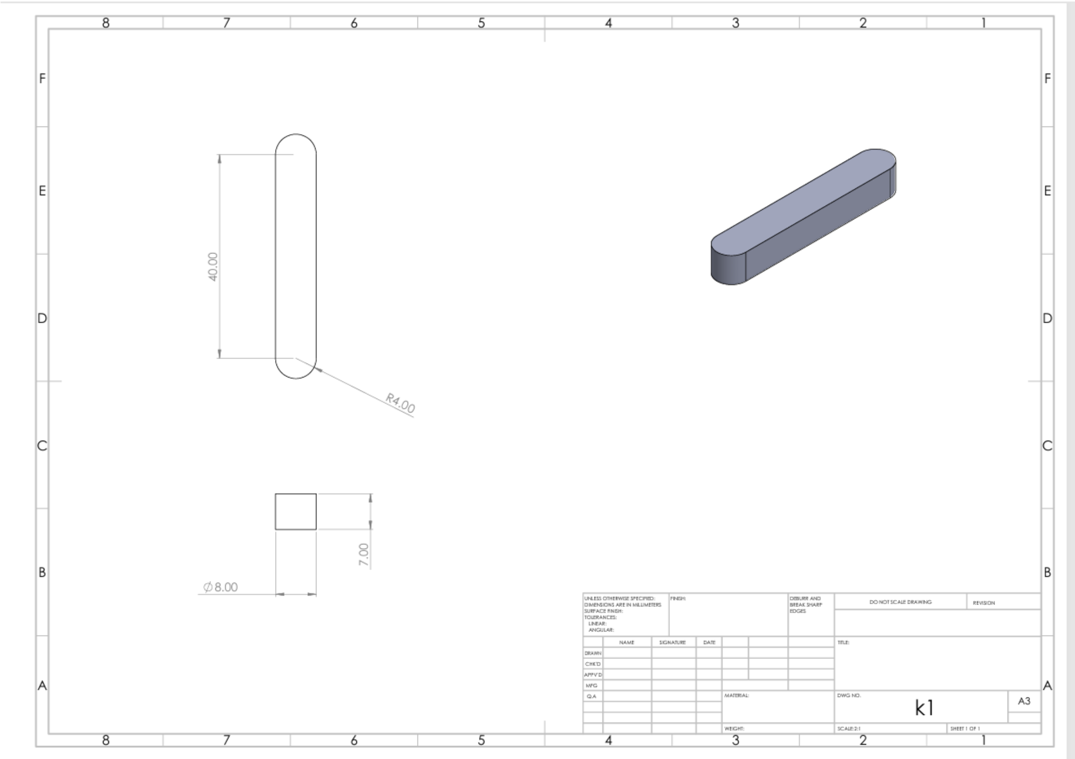

Figure 19:Designed key 1

The drawing of key 2 is shown below

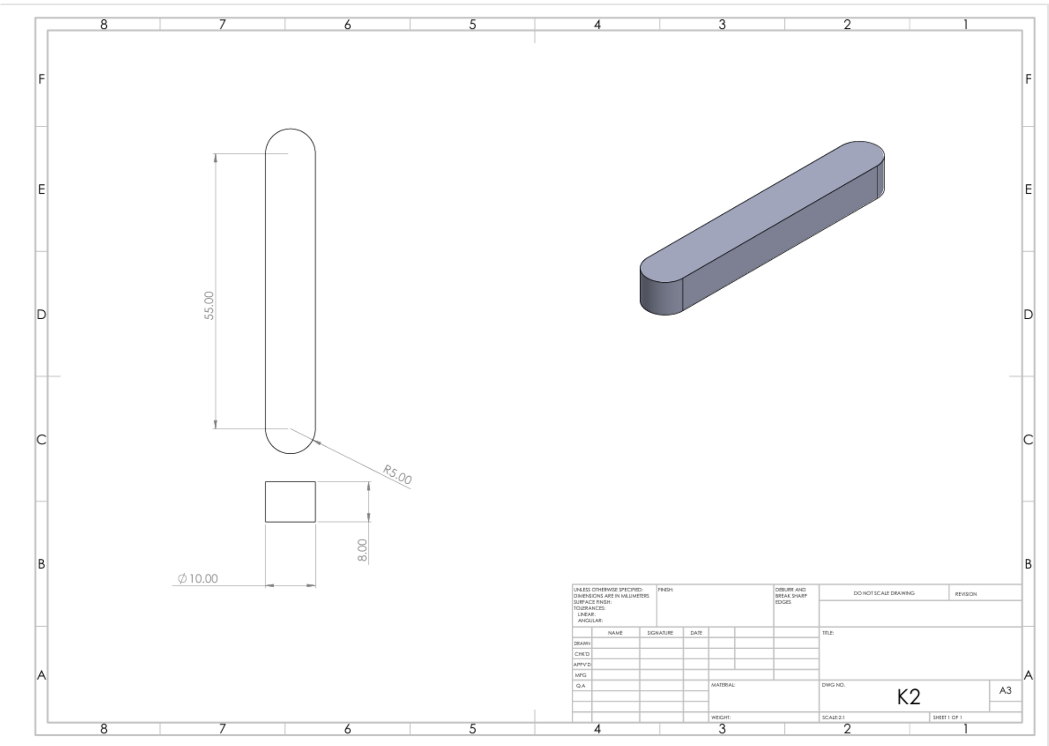

Figure 20:Designed key 2

The drawing of key 3 is shown below

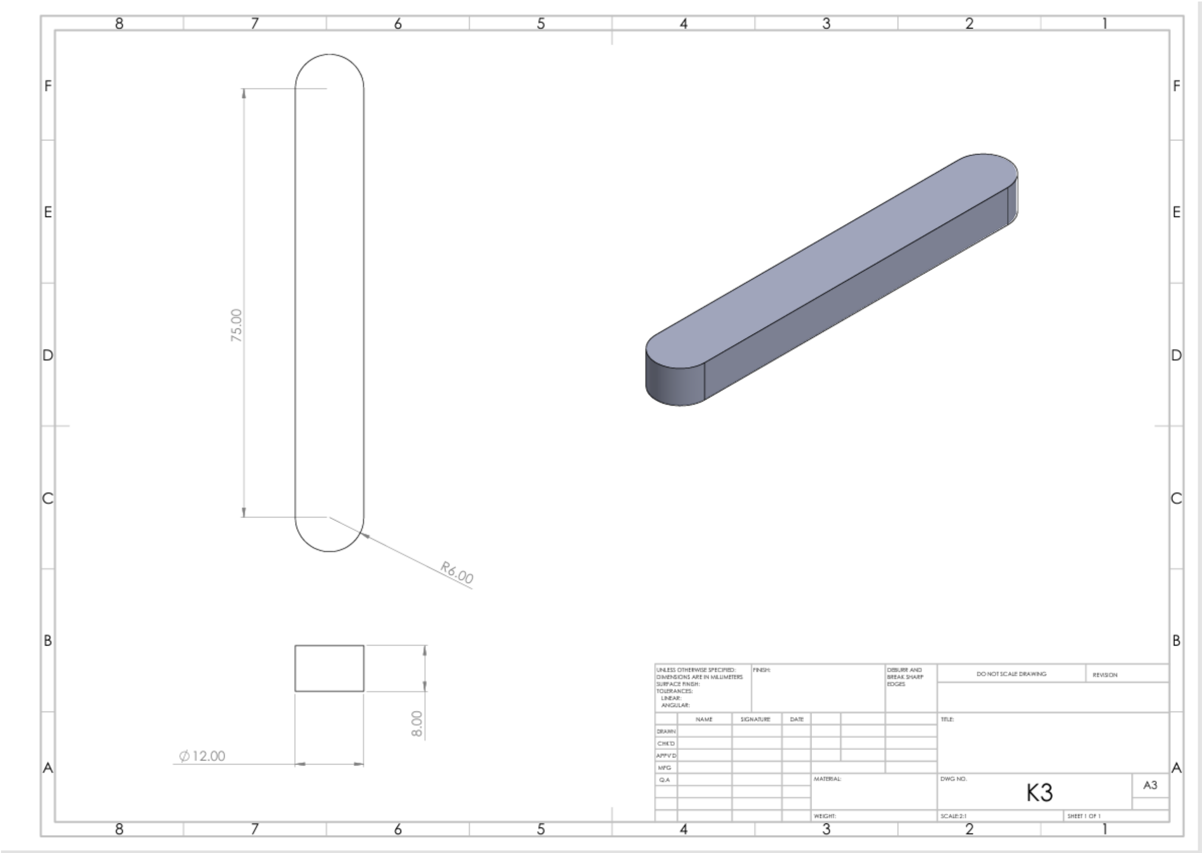

Figure 21:Designed key 3

4. Tabulated Results

The table below is the summary of the final gear design parameters after choosing the right module (m = 5 mm). It gives the number of teeth, the operating speeds, the transmitted loads, calculated stresses and factors of safety of each gear. The values ensure that all gears meet the strength criteria in the specified conditions of operation.

Table 7:Summary of gear design

Gear	**Teeth**	**Speed (rpm)**	**Module (mm)**	**Pitch Dia (mm)**	**Load (W_t) (N)**	**Stress (MPa)**	**FOS**	**Material**
2	20	1500	5	100	1591.5	33	7	C45 Steel
3	60	500	5	300	1591.5	30	8	C45 Steel

4	20	500	5	100	4775	~60	2.5	C45 Steel
5	100	100	5	500	4775	~50	3	C45 Steel

The table below presents the key design parameters for each shaft, including length, maximum bending moment, transmitted torque, and selected diameters. These values are obtained from bending moment and torsion analysis, ensuring that each shaft can safely withstand combined loading conditions.

Table 8:Summary of Shaft design

Shaft	Length (mm)	Max BM (Nmm)	Torque (Nmm)	Diameter (mm)	Material
Shaft 1	200	84,600	31,830	26	AISI 1045
Shaft 2	260	220,000	95,500	32	AISI 1045
Shaft 3	220	279,000	477,500	40	AISI 1045

The following table provides details of the selected rolling element bearings for each shaft. It contains bearing type, dimensions, equivalent dynamic load, and calculated life in hours. The chosen bearings meet the necessary life and reliability requirements to be used in continuous operation.

Table 9:Summary of Bearing design

Shaft	Bearing	Bore (mm)	OD (mm)	Width (mm)	Load P (N)	Life (hrs)
Shaft 1	6205	26	52	15	1270	14,800
Shaft 2	6206	30	62	16	2550	14,900
Shaft 3	6208	40	80	18	3810	89,600

The following table contains the dimensions and material choice of keys to connect gears to shafts. The sizes of the keys are determined according to the needs of the transmitted torques that have to be safe against failure modes of shear and crushing.

Table 10:Summary of key design

Shaft	Key Size (mm)	Length (mm)	Material	FOS
Shaft 1	8 × 7	40	Mild Steel	>2
Shaft 2	10 × 8	55	Mild Steel	>2
Shaft 3	12 × 8	75	Mild Steel	>2

The table shown below is a summary of the retaining ring dimensions and load carrying capacity. The components are applied to eliminate any axial movement of gears and bearings along the shaft and they are highly factor of safety designed.

Table 11:Summary of retaining ring design

Shaft	Thickness (mm)	Width (mm)	Axial Load (N)	Capacity (N)	FOS
Shaft 1	2	3	159	1800	11
Shaft 2	2.5	4	318	3000	9
Shaft 3	3	5	478	4500	9

The above tables collectively summarize the final design parameters of all major components, ensuring that the gearbox operates safely, efficiently, and within acceptable stress limits.

5. Conclusion

During this project, the entire design of a double reduction gearbox used in a conveyor belt system was completed considering the specified power and speed needs. The gear train was to be created to offer the necessary reduction in speed of 1:15 and be strong and durable. The first design considerations with a smaller module indicated that there were not enough factors of safety in the second stage gears and this prompted the choice of a bigger module ($m = 5$ mm) to enhance load carrying capacity. An extensive force analysis, stress analysis, and kinematic analysis were conducted on all gears. Shafts were developed with regard to combined bending and torsion with the influence of stress concentration created by keyways and shoulders. Bearings have been chosen according to the calculated loads and necessary life, which guarantees credible functioning. The keys and retaining rings were also designed such that they were able to transmit torque safely without causing the components to move vertically. The final design meets all the requirements of the functionality and strength with the right factors of safety so that the gearbox can work reliably and efficiently under the given conditions.

6. References

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